

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0380

Medical 138 kV
February 29, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0380. The Interconnection Customer proposes a 70.1 MW generating facility interconnecting at the Medical 138 kV substation in the Virginia area. The request was received on September 18, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 210 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$626,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Siuc to Fairland 138 KV line	Taber to Glenannan 345 KV	108.7	514	5.4
Adrian Energy Associates Llc 345/138kV Transformer Bank	Taber to Glenannan 345 KV	105.2	620	5.5
Taber 345 kV/138 kV Transformer	Taber to Dixleta 345 KV	104.9	918	5.9
Rustic Substation to Dixleta 345 KV	Huhson Substation to Dixleta Substation 345 KV	104.0	1,314	20.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Siuc to Fairland 138 KV line	Reconductor line	\$657,000	1% (\$9,000)
Adrian Energy Associates Llc 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$13,298,000	3% (\$417,000)
Taber 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$15,531,000	1% (\$105,000)
Rustic Substation to Dixleta 345 KV	Reconductor line	\$18,904,000	0% (\$95,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$626,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.